Electronic Instrumentation and Optoelectronics LAB

LAB2: Measuring with instruments. Systematic and random effects.

Instrument Drivers Tutorial

A instrument Driver is a set of functions (VI) that allow the control of a specific instrument without having to resort to the basic functions of the control interface functions or VISA. LabVIEW has a library on the functions panel for instrument drivers. This library must be created for each specific environment with the drivers of the instruments to be controlled. Controllers can be supplied by the manufacturer of the equipment or the manufacturer of the programming environment, in this case National Instruments through its web server (http \\ www.natinst.com) or CDROM.

External inteface model of a Instrument Driver

A LabVIEW instrument driver interacts with the rest of the system according to the general model in Figure 2.1. The working body is the set of functions that control the instrument. It consists of a set of VI that are developed with block diagrams so they can be analyzed and modified with tools of LabVIEW. The front panels of each VI (Interface development) can be used to interactively test and thus facilitate debugging programming errors..

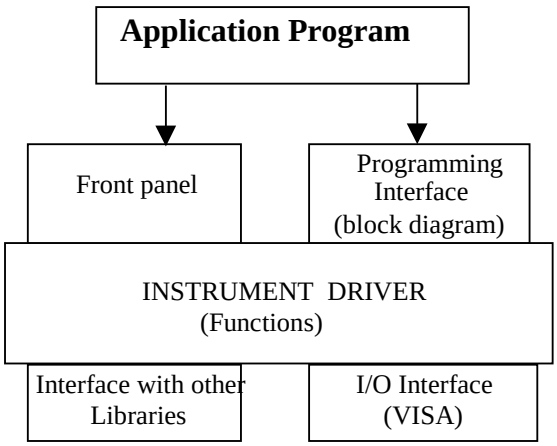


Fig. 2.1 Communication Model of a controller with its enviroment

The icons and connectors for each of the VI that form the driver are the inputs and outputs that connect the controller with the general program of the application. Thus the development of an application program will consist on (and modified if necessary) binding of the various extracted subVI from the instrument driver library and the auxiliary functions of processing and presentation of data.

The SubVI comprising a controller library are in turn formed by SubVI for the physical control of communication interface through VISA functions and other libraries of functions such as scanning and data processing.

Internal structure of the controllers (instrument divers)

The structure of a LabVIEW driver responds to the scheme in Figure 2.2. All instrument drivers would have to answer the same structure to facilitate program development. LabVIEW provides two types of functions for a controller: application functions and basic functions (Component functions). Application functions represent the highest hierarchical level of programming, allowing with a single VI the control of the functions and common measures of an instrument. The application functions are oriented to perform common actions almost immediately, but can also serve as an example for similar applications that suit the particular needs of the user

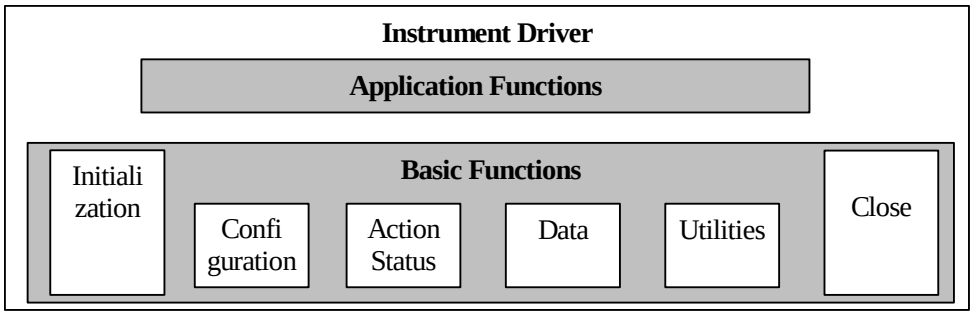


Fig. 2.2 Functional elements of a instrument driver

The basic control functions are a set of subVI that allow to program the different options for each instrument. The basic functions are divided into six categories: initialization, configuration, action / status, data, utilities, and close (close).

The initialization and closing funtions are mandatory for all drivers. The initialization function establishes the communication with the instrument and must be the first function to be executed. This function can also configure the instrument in a given initial state (default). The role of close function is to close the communication with the instrument and releasing system resources used to it.

The setup functions allow you to program the instrument in the most appropriate way to make a measurement or generate a signal later.

Action functions are those that cause the instrument to start or terminate an action, taking into account the configuration that was previously sent. These functions can include arming the trigger system or start an action (trigger). State functions allow us to know what is the status of the instrument and if there are pending execution operations.

The data functions are those that allow to transfer data from or to the instrument. They range from simple numeric data transmission to transmission of an entire vector of one or more dimensions.

The utility functions perform auxiliary functions not supported in the previous groups. For example resets, error handling, testing, calibration, etc.

In most professional instrument drivers it is included a VI (VI Tree) that displays all the functions that make up the controller and its hierarchy. It is also advisable to include the so-called "Starter VI" that serves to verify the functionality of our system without having to make any program and can serve as an example for future applications.